Silicon-Organic Hybrid (SOH) IQ Modulators with Sub-1 V π-Voltages Operating at 200 GBd 16QAM and 144 GBd 64QAM

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Abstract: We demonstrate compact high-speed silicon-organic hybrid (SOH) IQ modulators offering π-voltages below 800 mV. Our devices reach symbol rates of 200 GBd for 16QAM and of 144 GBd for 64QAM transmission over 85 km – record-high values for IQ modulators on the silicon photonic platform.

1. Introduction

The rapid scaling of computational workloads in data centers and artificial intelligence (AI) clusters leads to escalating bandwidth demands across all levels of optical networks. While massively parallel short-reach intensity-modulation-and-direct-detection (IMDD) links remain essential within racks and servers, the need for high-capacity connectivity across campus-scale distances (2 km to 20 km) becomes increasingly critical [1]. In this intermediate range, the performance of IMDD schemes suffers from channel impairments such as chromatic and polarization-mode dispersion or four-wave mixing, especially for high symbol rates [1]. Coherent transmission schemes, in contrast, can effectively compensate dispersion-related impairments while offering increased data rates per channel as well as enhanced receiver sensitivity and lower launch power, but rely on complex digital signal processing (DSP), which increases power consumption and latency. To address these challenges, optimized DSP architectures that reduce latency and power dissipation, commonly referred to as 'coherent lite', have emerged. Clearly, besides the DSP, the scalability of such approaches crucially relies on compact and efficient in-phase/quadrature modulators (IQM) that can be integrated at low cost on mainstream photonic platforms and that offer power-efficient operation at low drive voltages.

Over the previous years, various platforms for IQM have been explored and experimentally demonstrated. Among them, thin-film lithium niobate (TFLN) devices stand out due to their low optical loss and large bandwidths in excess of 100 GHz, permitting 16QAM and 32QAM transmission at symbol rates beyond 200 GBd [2]. However, the lengths of TFLN IQMs are typically of the order of 1 cm, dictated by the rather high π-voltage-length product $U_\pi L$ of typically 20 Vmm [2], which impedes integration into small form-factor packages. Moreover, TFLN is merely a standalone platform, which does not permit integration of additional circuitry, e.g., for coherent reception. In contrast to that, indium-phosphide- (InP-) based devices permit co-integration of transmitter and receiver circuitry [3] while offering bandwidths in excess of 100 GHz and symbol rates of 192 GBd [4]. However, InP devices still feature comparatively high $U_\pi L$ products in excess of 5 Vmm [5] and are subject to rather complex and costly fabrication processes on small-area substrates. Conversely, the silicon photonic (SiP) platform offers utmost scalability and fabrication yield [6] and has already been used for high-performance coherent receivers [7]. However, due to the absence of Pockels-type nonlinearities in bulk silicon, SiP IQM have to rely on carrier depletion in reverse-biased *p*-*n* junctions as a modulation mechanism, which is subject to intrinsic trade-offs regarding bandwidth, π-voltage, and optical loss. As a consequence, high-speed coherent communications with traveling-wave SiP IQM has so far been limited to symbol rates of 115 GBd or less [8], while the π-voltages may exceed 10 V and require power-hungry drive amplifiers and while the device lengths in excess of 4 mm are significantly larger than most other SiP devices. Smaller footprint, higher symbol rates up to 180 GBd, and drive voltages of 1.3 V have been achieved by dedicated microring-based IQM, but modulation was still limited to rather simple QPSK formats [9]. Similar restrictions apply to ultra-compact plasmonic devices, for which 16QAM transmission at symbol rates of 160 GBd have been reached [10].

In this paper we show that silicon-organic hybrid (SOH) integration [11] can overcome the stringent limitations of the SiP platform with respect to high-performance coherent transmission. We demonstrate a high-speed IQM with π-voltages below 800 mV, reaching symbol rates of 200 GBd for 16QAM and of 144 GBd for 64QAM transmission – a twofold increase compared to earlier demonstrations [12]. In a proof-of-concept experiment, we operate the IQM directly with sub-1.1 V signals obtained from a high-speed waveform generator, without any additional RF amplifiers. We perform a systematic sweep of the symbol rate and demonstrate achievable information rates (AIR) of 712 Gbit/s for 16QAM signals at 192 GBd and 743 Gbit/s for 64QAM signals at 144 GBd transmitted over 85 km of standard single-mode fiber (SMF). To the best of our knowledge, our experiments demonstrate the highest symbol rate so far achieved in a coherent data transmission experiment using IQM on the SiP platform.

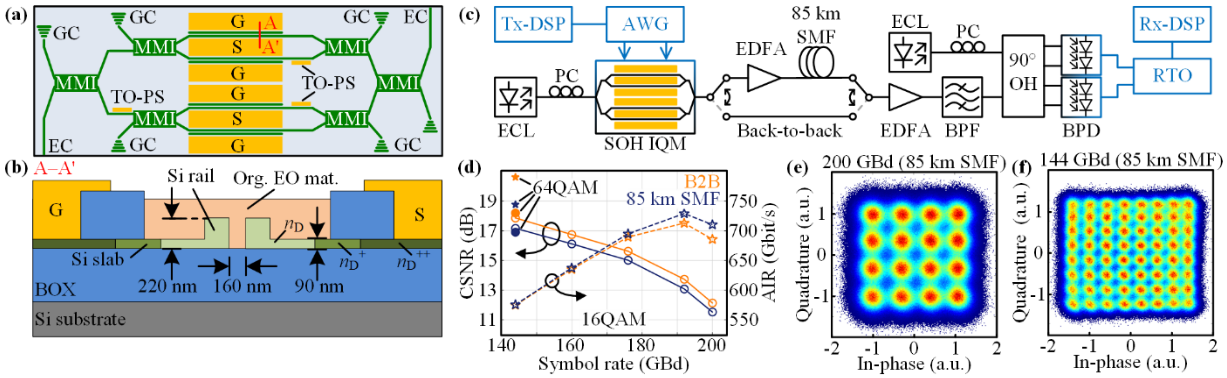

Fig. 1. Silicon-organic hybrid (SOH) IQ-modulator principle and high-speed transmission demonstration. **(a)** Top-view schematic of SOH IQ-modulator, consisting of two nested Mach-Zehnder modulators with electrodes (yellow) in ground-signal-ground (GSG) configuration and thermo-optic phase shifters (TO-PS). Optical waveguides, multimode interference couplers (MMI), and grating couplers (GC) or edge couplers (EC) are shown in green. **(b)** Cross-section of the SOH slot-waveguide phase shifter along the line A–A' as defined in Subfigure (a). The slot, defined by a pair of Si rails, is filled with an organic EO material (org. EO mat.). The doping concentration (n_D, n_D^+, n_D^{++}) of the Si slab, connecting the rails to the ground (G) and signal (S) electrodes, increases with distance from the rails. **(c)** Measurement setup for coherent communication experiments: An external-cavity laser (ECL) provides the carrier via a polarization controller (PC) to the SOH IQM, which is driven by an arbitrary-waveform generator (AWG). Digital signal processing (Tx-DSP) compensates the RF losses between the AWG and the IQM. The modulated light is either amplified by an erbium-doped fiber amplifier (EDFA) and transmitted over 85 km of standard single-mode fiber (SMF), or it is sent directly to another EDFA (back-to-back, B2B). A subsequent bandpass-filter (BPF) suppresses out-of-band noise before the signal is fed to a coherent receiver, consisting of a 90°-optical hybrid (90° OH) and a pair of balanced photodiodes (BPD). The BPD outputs are recorded by a real-time oscilloscope (RTO). Offline digital signal processing (Rx-DSP) is used to recover the data and to calculate the constellation signal-to-noise ratio (CSNR). **(d)** CSNR (circles, solid lines) and achievable information rate (AIR, asterisk, dashed lines) as a function of symbol rate for 16-state quadrature-amplitude modulated (16QAM, unfilled) and 64-state QAM (64QAM, filled) signals as transmitted over a SMF (blue), or in B2B configuration (orange). Constellation diagrams **(e)** of the received 16QAM signal at 200 GBd, and **(f)** 64QAM signal at 144 GBd after the SMF.

2. Silicon-organic hybrid (SOH) IQ-modulator

A simplified top-view schematic of our SOH IQM is shown in Fig. 1(a). Optical waveguides are shown in green and metal electrodes in yellow. Grating couplers (GC) or edge couplers (EC) are used to couple light between SMF and on-chip strip waveguides. The IQM consists of two nested Mach-Zehnder modulators (MZM), where 2×2 multi-mode interference couplers (MMI) are used for splitting and combining the light. Three thermo-optic phase shifters (TO-PS) are used to set the operating point of the two individual MZM and the phase between the in-phase (I) and quadrature (Q) MZM. Each MZM comprises a coplanar waveguide (CPW) in a ground-signal-ground (GSG) configuration acting as a travelling-wave RF transmission line to guide the modulating electric field. In-between the signal and the ground electrodes, 1 mm long SOH slot waveguide phase shifters modulate the light [11]. Strip-to-slot waveguide converters (not shown) are used at the input and output of each phase-shifter [13].

A cross-section of a single phase shifter along the line A–A' in Fig. 1(a) is depicted in Fig. 1(b). The slot waveguide consists of two silicon rails, confining the optical mode within the slot filled with an organic electro-optic (EO) cladding material (SOXD123, SilOriX GmbH, Germany). The organic EO material is dispensed in a back-end-of-line process and subsequently poled by applying an electric field at elevated temperatures, see [11] for details. Efficient modulation requires a strong spatial overlap between the light and the modulating RF field. The aluminum electrodes are connected to the Si rails via metal vias and thin Si slabs, leading to a strong confinement the RF field to the slot. The resistivity of the doped Si slabs and the capacitance of the silicon rails represent a low-pass filter, limiting the modulation bandwidth. An advanced doping scheme reduces the slab resistivity while keeping the optical loss low: The Si rails and their adjacent slab regions have a low donor concentration (n_D) to minimize free-carrier absorption, while the doping concentration increases (n_D^+) further away from the waveguide and reaches a highly degenerate doping (n_D^{++}) at the ohmic contacts, see [14] for details. Characteristic parameters of our IQM are the optical fiber-to-fiber transmission (17.7 dB at a wavelength of 1550 nm), and the half-wave voltage of the nested MZM, which amounts to 780 mV and 790 mV for the two arms.

3. High-speed coherent communication experiment

We demonstrate the performance of our SOH IQM in a coherent high-speed transmission experiment. The setup is illustrated in Fig. 1(c). A tunable external-cavity laser (ECL, 1550 nm, 17.8 dBm) provides the optical carrier. The electrical drive signals for the in-phase (I) and quadrature (Q) channels are generated by a high-speed arbitrary-waveform generator (AWG, M8199B, Keysight) and fed to the IQM via 20 cm-long RF cables and a double GSG RF probe. With offline DSP (Tx-DSP), we synthesize differently formatted electrical modulation signals at various symbol rates. The Tx-DSP comprises the generation of pseudo-random bit sequences (PRBS), pulse-shaping with a root-raised cosine (RRC) filter (roll-off $\beta = 0.05$), and a predistortion to compensate the frequency-dependent RF characteristic of the AWG and the cables, but not of the modulator. Clipping of the signal reduces the peak-to-average power ratio to 9 dB. Both MZM are terminated via a second RF-probe and coaxial 50 Ω resistors. The modulated signal is amplified by an erbium-doped fiber amplifier (EDFA) and transmitted over 85 km of standard SMF. For back-to-back (B2B) measurements, the EDFA and SMF are bypassed. A second EDFA adjusts the power to 2 dBm before feeding the signal to a 90° optical hybrid (90°-OH). A second ECL

(1550 nm, 17.8 dBm) provides the local oscillator. The outputs of the 90°-OH are detected by balanced photodiodes, recorded by a high-speed real-time oscilloscope (RTO, UXR 1004A Keysight) and processed offline (Rx-DSP). The Rx-DSP comprises up-sampling to two samples per symbol, chromatic dispersion compensation, timing recovery, linear equalization, frequency and phase offset compensation, and additional post-equalization based on a decision-directed least-mean-square algorithm.

In our experiments, we transmit 16QAM signals via 85 km of SMF and in a B2B configuration. Symbol rates range from 144 GBd to 200 GBd, and the peak-to-peak voltage swing of the drive signal is below 1.1 V. The calculated constellation signal-to-noise ratios (CSNR) are depicted in Fig. 1(d), indicated by circles and solid lines. For 16QAM (unfilled circles) transmission over 85 km of SMF (blue), the CSNR decreases from 17.2 dB at 144 GBd to 11.5 dB at 200 GBd with a penalty of only 0.7 dB compared to the B2B measurement (orange). Furthermore, we estimate the generalized mutual information (GMI), assuming a channel model with additive white Gaussian noise [15]. For 200 GBd, we calculate an NGMI of 0.857 which is above the threshold for SD-FEC with 25 % overhead [16]. The corresponding constellation diagram is shown in Fig 1(e). For bit-interleaved coded modulation, the GMI quantifies the achievable information rate (AIR), indicated by asterisks and dashed lines in Fig. 1(d). The highest AIR is 712 Gbit/s for 16QAM at 192 GBd, which is on par with the single-polarization AIR obtained for 16QAM signaling using best-in-class high-speed TFLN modulators [2]. Note that slot-waveguide base structures of the SOH devices have been obtained through and industry-standard silicon photonic foundry process and are seamlessly co-integrated with standard SiP circuitry [6]. The approach is hence ideally suited for implementing coherent transceiver PICs that combine IQM and IQ receivers on a common die. Note also that SOH devices have been shown to exhibit long-term photostability of more than 2000 h provided that suitable encapsulation techniques are used to prevent in-diffusion of ambient oxygen [17]. We also tested our device at high-order modulation formats, transmitting 64QAM signals at 144 GBd and an NGMI above SD-FEC threshold (25 % overhead) and offering an AIR of 743 Gbit/s, see Fig. 1(f). To the best of our knowledge, both the 200 GBd 16QAM and the 144 GBd 64QAM transmission represent the highest symbol rates so far achieved for the respective modulation format by an IQM on the SiP platform. Compared to earlier demonstrations of SOH IQMs [12], we could increase the symbol rate by a factor of two.

4. Summary

We demonstrate a compact high-speed silicon-organic hybrid (SOH) IQ modulator (IQM) that offers π-voltages below 800 mV and that can be seamlessly co-integrated with standard silicon photonic circuitry. We demonstrate the viability of the device by transmitting 16QAM signals at a symbol rate of 200 GBd and 64QAM signals at a symbol rate of 144 GBd, both with an NGMI above the soft-decision forward-error-correction (SD-FEC) threshold with 25 % overhead. To the best of our knowledge, these symbol rates represent the highest values so far achieved for the respective modulation format by an IQM on the SiP platform. Our devices reach single-polarization achievable information rates (AIR) of 712 Gbit/s for 192 GBd 16QAM and of 743 Gbit/s for 144 GBd 64QAM. The high efficiency of the SOH modulators enables operation at peak-to-peak voltage swings below 1.1 V, thereby avoiding power-hungry RF drive amplifiers. Our results show the immense potential of the SOH platform for scalable and highly efficient coherent transceivers that may be used for campus-scale distances (2 km to 20 km).

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5. References

[1] X. Zhou *et al.*, "State-of-the-Art 800G/1.6T Datacom Interconnects and Outlook for 3.2T," in Proc. *OFC* 2023, Paper W3D.1.

[2] S. Almonacil *et al.*, "260-GBaud Single-Wavelength Coherent Transmission Over 100-km SSMF Based on Novel Arbitrary Waveform Generator and Thin-Film Niobate I/Q Modulator," *J. Lightw. Technol.*, 41 (12), 3674-3679 (2023).

[3] M. Ziari *et al.*, "High-performance 100 Gbaud Coherent Photonic Modules," in Proc. *OFC 2021*, Paper F2A.3.

[4] H. Wakita *et al.*, "InP-Based on-Board Coherent Tx Front-End With 100-GHz Bandwidth Utilizing Flip-Chip Technology Enabling 2-Tb/s/λ Optical Transmission," *J. Lightw. Technol.*, 43 (4), 1819-1825 (2024).

[5] Y. Ogiso *et al.*, "High Speed InP Modulator for Beyond 200 Gbaud," in Proc. *OFC 2024*, Paper Tu2D.7.

[6] S. Y. Siew *et al.*, "Review of Silicon Photonics Technology and Platform Development," *J. Lightw. Technol.*, 39 (13), 4374-4389 (2021).

[7] T. Richter *et al.*, "1 Tb/s and 800 Gb/s real-time transmission at 138 GBd over a deployed ROADM network with live traffic," in Proc. *OFC 2023*, Paper Th4C.1.

[8] E. Berikaa *et al.*, "Silicon Photonic Single-Segment IQ Modulator for Net 1 Tbps/λ Transmission Using All-Electronic Equalization," *J. Lightw. Technol.*, 41 (4), 1192-1199 (2023).

[9] A. Geravand *et al.*, "Ultrafast coherent dynamics of microring modulators," *Nature Photon.*, 19, 740–750 (2025).

[10] L. Kulmer *et al.*, "256 GBd single-carrier transmission over 100km SSMF by a plasmonic IQ modulator," in Proc. *ECOC 2023*, 1449-1452.

[11] W. Freude *et al.*, "High-Performance Modulators Employing Organic Electro-Optic Materials on the Silicon Platform," *IEEE J. Sel. Top. Quantum Electron.*, 30 (4), 1-22 (2024). Corrections to Table II: *IEEE J. Sel. Top. Quantum Electron.* 31 (2), Art. no. 9800101 (2025).

[12] S. Wolf *et al.*, "Coherent modulation up to 100 GBd 16QAM using silicon-organic hybrid (SOH) devices," *Opt. Express*, 26 (1), 220-232 (2018).

[13] R. Palmer *et al.*, "Low-Loss Silicon Strip-to-Slot Mode Converters," *IEEE Photon. J.*, 5 (1), Art. no. 2200409 (2013).

[14] H. Zwickel *et al.*, "Verified equivalent-circuit model for slot-waveguide modulators," *Opt. Express*, 28 (9), 12951-12976 (2020).

[15] M. Ivanov et al., "On the Information Loss of the Max-Log Approx-imation in BICM Systems," *IEEE Trans. Info. Theory*, 62 (6), 3011-3025 (2016).

[16] A. Graell I Amat *et al.*, "Forward Error Correction for Optical Transponders," in *Springer Handbook of Optical Networks*, B. Mukherjee *et al.*, eds. (Springer, Cham, 2020).

[17] A. Mertens *et al.*, "Silicon-organic hybrid slot waveguide modulators on the verge of industrial adoption," *PIC magazine* 3, 18–23 (2024).